

Coherence in Biology

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The great success of molecular biology arises from the establishment of the atomic structure of biological systems such as DNA or proteins. The activity of these systems does, however, not follow in a simple way from structure as it frequently can be switched on or off. From the point of view of physics this must be expressed in terms of non-linear excitations. Quite different types of excitation often have common general features which has given arise to Haken's synergetics [1]. Establishment and maintenance of such excitations requires the supply of energy. Energy supply, in general, leads to heating. In cases which are of biological interest, however, metabolic energy supply leads to the establishment of organisation, Prigogines's dissipative structures [2]. Whether the one or the other holds must be investigated in detail for each system. No general rule has been found, so far, which would permit a decision between the two possibilities from structure only.

About fifteen years ago it has been conjectured that coherent excitations should play an important role in biological activity; and the conjecture was supported by model calculations ([3], [4], [5]). Such excitations necessarily contain essential non-linear features. In contrast to the realm of linear excitations, systematic investigation then becomes impossible, in principle, as the number of states is too large [6]. Biological systems have, however, developed to fulfil a certain purpose, and it is permissible, therefore, to ask for the purpose of a certain excitation, a question which in physics is permissible only when dealing with machines or similar constructions. Special assumptions must thus be made, and tested against experiment.

The following gives a short survey on the predictions of the theory. The subsequent papers discuss experimental evidence. Coherence determines properties at a space-time point (x^1, t^1) when they are known at another (x, t) , such as phase and amplitude of coherent waves. Three basic types of coherent excitations have been proposed:

- A. Coherent excitation of a single polar mode ([5], III C).
- B. Excitation of a metastable highly polar state ([5], III B).
- C. Vibrations arising from more complex processes and giving rise to limit cycles, or Lotka Volterra oscillations ([5], III E.).

Certain material conditions must be satisfied to permit such excitations. Coherent excitation of a single mode (A) of the band of polar modes may arise from random energy supply at a rate S to all or only some of these modes, provided S exceeds a critical S_0 , $S > S_0$, and provided these modes are in strong non-linear, interaction with a heat bath (e.g. cell water) kept at constant temperature. Establishment of the coherent excitation will require a certain time after the start of energy supply. It should be minimal if the energy is supplied directly to the mode that will be excited coherently. The system, thus, possesses storing capability. A range of possible frequencies has been proposed (cf [5], III A), in particular the region of 10^{11} Hz (mm wave region) for sections of membranes, but also lower as well as considerably higher frequencies characteristic for various biological molecules. Furthermore, plasma oscillations of electrons in conduction bands may have to be considered.

A metastable highly polar state (B) may arise from the shape dependence of the electrostatic and elastic energies of a homogeneously polarised and deformed system. This possibility should in particular arise in highly polarisable and deformable materials like proteins. This excitation may be supplemented by small conformational changes, such as the transfer of proteins to different positions, e.g. as discussed below by Genzel. It must be realised in this context that the electrostatic energy of a highly polarised molecule may be considerable and can overwhelm changes in local binding energies. The high electric field in a membrane may be expected to lift proteins dissolved in it into their metastable highly polar state. When in the cytoplasm, on the other hand, the high field due to the polarity of a molecule in the metastable state is likely to be screened by counterions.

The two excitations A and B may have far reaching consequences on the activity of the relevant systems. It may be noted in particular that they can be switched on, or off, depending on the energy supply, giving rise to a particular activation or deactivation. Oscillations of type A yield long range, frequency selective, interactions between systems with equal excitation frequencies; it will be noted that electrostatic interaction is usually screened by counterions. This frequency selective interaction may lead to long range attraction. If this takes place between enzymes and substrates then it proceeds and initiates the usual short range chemical interaction by bringing relevant systems together. We note here a multicausal process, typical for many biological events. It has also been suggested that the selective long range interaction may be relevant for the control of cell division, important in the cancer problem ([5], IV D). Excitation A may also provide facilities for communication within and between cells.

It has been proposed [7] that the high field in membranes would require the use of non-linear optics for the vibrations based on it and hence give rise to self focusing and to involvement in the filamentous microstructure observed in eucaryotic cells.

The highly polar metastable state (B), [8, 9], could represent the active state of enzymes. It stores energy, and, through its internal field can reduce activation energies. This excitation could also be effective in transporting energy from protein to protein. In an array of proteins dissolved in membranes in this way it would act as a large soliton and it has been shown, in fact, that the relevant basic potential energy also leads to ordinary soliton solutions [10].

Considering an array of localised enzymes with activated state described by B, when excitation A attracts substrates, leads to a Lotka-Volterra type of periodic enzyme reaction. Taking account of enzyme-enzyme interaction then yields limit cycles, excitation C. The periodic enzyme excitation - diexcitation in view of the high polarity of B, then causes electric vibrations accompanying the limit cycle, and a sensitivity to weak external fields. This may trigger a collapse of the limit cycle and hence a sudden liberation of considerable energy, as will be reported below by Kaiser.

The consequences of the excitations described above must have decisive influence on the activity of biological systems. They thus might form the dynamic complement to structure. A series of experiments has been designed to establish the existence of these excitations rather than with their biological significance. For excitation of coherent vibrations (A) the most direct method, at first sight, appears to be laser Raman effect when the frequency is above 200 cm^{-1} . For then the ratio of anti-Stokes to Stokes intensity must be noticeably larger than in thermal equilibrium. Such experiments have, so far, been carried out on bacterial cells only where a number of experimental difficulties arise. Some of them are connected with the high dilution required to keep the cells in their active state so that a very high degree of synchronisation is necessary (cf [11]). These difficulties have been overcome in the experiment reported by Drissler, below, which indicates strong excitation of a vibration which attracts nutrient molecules into a cell. This interpretation would permit verification by direct measurement of the nutrient influx into a cell which should exceed that arising from diffusion.

Existence of a metastable state (B) in Langmuir-Blodgett layers of proteins is reported below by Hasted. It is reached through application of very high electric fields which indicates implication of the required high dipole moment though at a later stage this will be screened by counterions. Investigations by Mascarenhas [12] with the electret method may be relevant in this context.

Decisive evidence on A arises from experiments on the action of low intensity mm waves on biological materials. It will be noticed that the frequencies are in

the range predicted for membrane vibrations. Intensities usually are below 10mW/cm^2 which excludes thermal effects, as well as direct i.e. non-linear action on an ordinary system [13]. The conclusion must thus be drawn that biological systems have developed an organisation that makes them particularly sensitive to low frequency mm waves. Note that a properly tuned radio is, of course, very sensitive to radio waves of extremely low intensity. This instrument has been specially designed for our needs. There is no evolutionary need, however, for biological systems to be sensitive to external mm waves. The conclusion must then be drawn that the biological systems themselves make use of vibrations in this frequency region i.e. that they use excitation A, or possibly C.

A basic result of the theory requires the energy supplied to cause the excitation to exceed a critical value S_0 . This gives rise to a step-like response similar to a phase transition. If the system is in a state close to the step this small external energy supply may trigger off the excitation. Also if the coherent excitation is already excited, but not to its ultimate value, then energy supply at the frequency in question will increase the amplitude. The exact type of excitation may, of course, vary from case to case. Thus apart from the above mentioned case A, a limit cycle of Type C may also be sensitive to weak external fields, as mentioned above.

Sensitivity to low intensity radiation may also arise from cooperative action of a number of charges as is available in the highly polar metastable state B.

The subsequent articles by Grundler, Kremer and Nimtz provide excellent evidence for the existence of such excitations though they can not yet specify details. More direct evidence for the use made of coherence in biological activity requires investigation of specific biological processes. A first step in this direction is discussed below, by Kell. Evidence for attraction between cells, arising from excitation of membrane vibrations has been obtained for the case of red blood cells by Rowlands, as shown in his article, below. Experiments by Brewer and Bell [14] on the long range interaction between amoebae and anion-exchange particles, and by Roberts et. al. [15] on the intracellular migration of nuclear proteins might also find their explanation in terms of our long range frequency selective forces. Evidence for the existence of relatively low frequency vibrations, with possible significance for cell division is presented below by Pohl. It thus appears that the evidence for the existence of coherent excitations is very strong. So far, however, very little insight has been gained on their specific biological significance. Experiments of a different nature will have to be designed for this purpose. We are dealing with cooperative phenomena, and it would seem that multicellular systems should make use of coherent excitations to a larger extent than single cells. Investigations of differentiated tissues are thus highly desirable.

References

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P.S.: To avoid unnecessary experimentation: Macroscopically, excitation A will be highly multipolar and hence does not lead to energetically unfavourable emission of radiation.